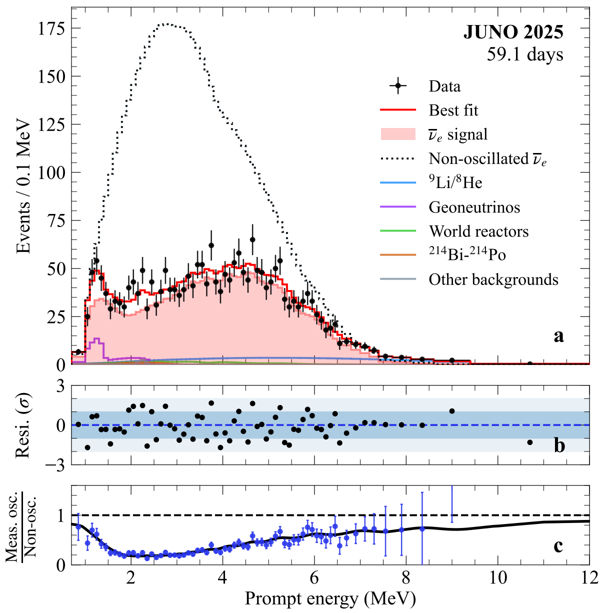

Fig. 3 Measured energy spectrum of prompt IBD candidates. **a**, Black points show measured data with statistical error bars, with the red curve indicating the best-fit oscillation model. Shaded red region represents expected antineutrino signal. Black dotted line represents non-oscillated reactor neutrino expectation. Backgrounds are indicated by other solid color lines. **b**, Residuals quantifying the statistical consistency between data and the complete model. **c**, Ratio of the measured oscillated spectrum to non-oscillated prediction.

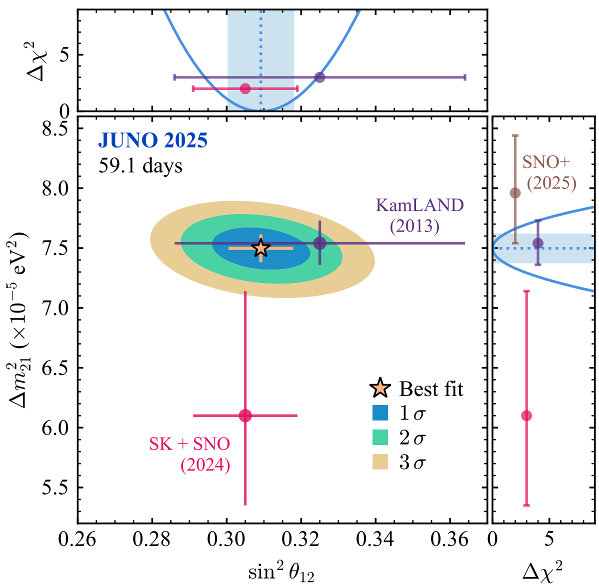

Fig. 4 Results on solar oscillation parameters. Confidence intervals of $\sin^2\theta_{12}$ and Δm^2_{21} from a spectral fit of measured prompt antineutrino energy spectrum shown in Fig. 3. Shaded elliptically-shaped areas correspond to 1σ, 2σ, and 3σ confidence levels. The upper panel provides the one dimensional $\Delta\chi^2$ for $\sin^2\theta_{12}$ obtained by profiling Δm^2_{21} (blue line) and the blue shaded region as the corresponding 1σ interval. The right panel is the same, but for Δm^2_{21}, with $\sin^2\theta_{12}$ profiled. The star marks the best fit values of JUNO, and the error bars show their one-dimensional 1σ confidence intervals. Results from other measurements of reactor neutrinos (KamLAND [13] and SNO+ [54]) and solar neutrinos (combined Super-Kamiokande (SK)+SNO [61]) are shown for comparison.

the Daya Bay experiment [55, 56], plus a subdominant term covering their differences, including contributions from Taishan, Yangjiang, and other distant reactors. Minor corrections from spent nuclear fuel and non-equilibrium isotopes were also included. More details are given in Methods.

Spectral fit and results

The observed prompt energy spectrum of IBD candidates, shown in Fig. 3, was compared with a detailed prediction to determine the neutrino oscillation parameters. This prediction incorporated the complete sequence of effects from reactor neutrino production to signal detection (details in Methods): reactor fluxes evaluated via a semi-model-independent combination of Daya Bay near detector data [55, 56] and theoretical models [57, 58], baselines from JUNO site to multiple reactors as provided in Ref. [5], time-dependent fission fractions, neutrino oscillation including Earth's matter effects [59, 60], interaction cross section [40, 41], detector response, and estimated backgrounds (Table 1).

Systematic uncertainties in the reactor flux prediction, shown in Table 2, are dominated by the 1.2% contribution from the Daya Bay reference spectrum [56], followed by the 1% uncertainty in the JUNO target proton number, which is driven by the liquid scintillator volume and hydrogen fraction.

The spectral distortion observed in the prompt-energy spectrum in Fig. 3 was primarily shaped by solar oscillation parameters Δm^2_{21} and $\sin^2\theta_{12}$, with a weaker dependence on Δm^2_{31}. Due to the currently limited statistics, our oscillation analysis concentrated on determining the